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Highly enantioselective catalytic isopropenylation of aldehydes using diisopropenylzinc

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Abstract

Diisopropenylzinc adds highly enantioselectively to various aldehydes in the presence of a catalytic amount of chiral amino alcohols to give enantiomerically enriched allyl alkanols with up to 92% e.e. © 1999 Elsevier Science S.A. All rights reserved.

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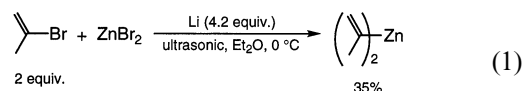
1. Introduction

Enantioselective alkylation of aldehydes is an important method for the preparation of chiral secondary alcohols [1]. We and other groups have examined comprehensively the enantioselective addition of dialkylzincs to aldehydes using chiral amino alcohols as chiral catalysts [2]. However, in most cases, only di(*n*-alkyl)zincs such as diethylzinc have been utilized. A limited number of papers has appeared on the use of diorganozincs with α -branched alkyl substituents (isopropyl [3] and cyclopropyl [4]). We [5] and Oppolzer et al. [6] reported enantioselective alkenylations of aldehydes using alkenylethylzinc which does not possess any α -branched alkenyl substituent.

We disclose here a novel way of catalytic and enantioselective isopropenylation of aldehydes using diisopropenylzinc possessing α -branched structure.

2. Results and discussion

Diisopropenylzinc was prepared by the reaction between zinc bromide, isopropenyl bromide and lithium under ultrasonic condition [4,7]. Then diisopropenylzinc was purified by sublimation (Eq. (1)).



At first, an enantioselective isopropenylation of benzaldehyde was examined using (*S*)-diphenyl(1-methylpyrrolidin-2-yl)methanol (DPMPM) **1** [8] as a chiral catalyst (Table 1, Entries 1–4). In the presence of 5 mol% of (*S*)-**1** in toluene at 0°C, asymmetric isopropenylation proceeded to give (*S*)-allyl alcohol **3a** in high yield (91%) and optical yield (89% e.e.) (Entry 1). When the reaction was performed at –10°C, both the yield and e.e. were slightly decreased (Entry 2). The yield was improved when hexane was used as a solvent instead of toluene (Entry 3). An increase in the amount of the chiral catalyst **1** (15 mol%) resulted in a slight increase in e.e. (Entry 4).

Next, *N,N*-dialkylnorephedrine were utilized as chiral catalysts (Entries 5–7). The enantioselectivity was effected by the structure of substituents on the nitrogen atom of norephedrine. Among *N,N*-diisobutylnorephedrine (DIBNE **2**) [9], *N,N*-dipropylnorephedrine (DPNE) [3,9], and 2-pyrrolidinyl-1-phenylpropan-1-ol (PPP) [3,9,10], DIBNE **2** was found to be the best chiral catalyst for the asymmetric isopropenylation of aldehydes.

Various aldehydes were submitted to the enantioselective isopropenylation using (*S*)-DPMPM **1** (5 mol%) or DIBNE **2** (20 mol%) (Table 2). Except for asymmet-

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Table 1
Enantioselective isopropenylation of benzaldehyde under various reaction conditions

Entry ^a	Chiral catalyst	(mol%)	Time (h)	Yield (%)	E.e. (%) ^b
1	(<i>S</i>)-DPMPM 1	(5)	2	91	89
2 ^c	(<i>S</i>)-DPMPM 1	(5)	4.5	85	85
3 ^d	(<i>S</i>)-DPMPM 1	(5)	2	95	86
4	(<i>S</i>)-DPMPM 1	(15)	2	85	90
5	(1 <i>S</i> ,2 <i>R</i>)-DIBNE 2	(20)	2	73	88
6	(1 <i>S</i> ,2 <i>R</i>)-DPNE	(20)	2.3	71	85
7	(1 <i>S</i> ,2 <i>R</i>)-PPP	(20)	2	90	80

^a Unless otherwise noted, the reactions were performed in toluene at 0°C. The molar ratio of aldehyde:diisopropenylzinc was 1.0:3.0.

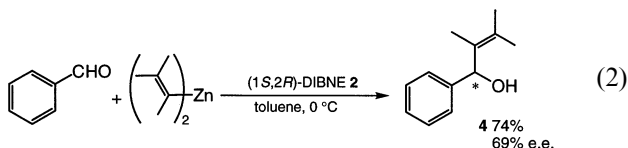
^b E.e. was determined by HPLC using a chiral column.

^c The reaction was performed at –10°C.

^d Hexane was used as a solvent.

ric isopropenylation of benzaldehyde, DIBNE **2** was proved to be a chiral catalyst with higher enantioselectivity compared with DPMPM **1** (Entries 3–6). In the case of 4-methoxybenzaldehyde, the e.e. reached the highest value of 92% (Entry 6). An unstable diallyl alcohol **3e** was obtained in 73% e.e. by asymmetric diallyl alkenylation of cinnamaldehyde (Entry 8). Also, an aliphatic aldehyde was enantioselectively isopropenylated under the same reaction conditions with 73% e.e. by use of DIBNE **2** (Entry 9).

We also prepared bis(1,2-dimethyl-1-propenyl)zinc by the same manner as diisopropenylzinc. It was also found that bis(1,2-dimethyl-1-propenyl)zinc enantioselectively adds to benzaldehyde by using (1*S*,2*R*)-DIBNE **2** as a chiral catalyst to give the corresponding allyl alcohol (+)-**4** with 69% e.e. in 74% (Eq. (2)).



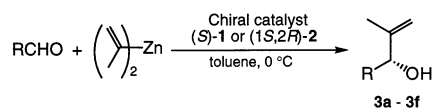
As described, we developed the first highly enantioselective isopropenylation [11] of aldehydes by diisopropenylzinc in the presence of a catalytic amount of chiral amino alcohols DPMPM **1** or DIBNE **2**. Various enantiomerically enriched isopropenyl alcohols obtained by the present reaction are synthetically very useful compounds [12].

3. Experimental

General: optical rotation was measured by a Jasco DIP-1000 polarimeter. IR spectra were recorded with a Horiba FT210 spectrophotometer. ¹H-NMR spectra were measured with a Bruker DPX300 spectrometer using tetramethylsilane as an internal standard and CDCl₃ was used as solvent. High resolution mass spectra (HRMS) were obtained with a Jeol JMS-SX102A mass spectrometer. Diethyl ether was distilled from lithium aluminum hydride (LAH) and dried over molecular sieves 4 Å (MS 4 Å). Toluene was distilled from calcium hydride and dried over MS 4 Å.

Preparation of a toluene solution of diisopropenylzinc [4,7]: all operations were performed under an argon atmosphere. Zinc bromide (16 g, 71.1 mmol) was heated in vacuo then cooled to room temperature. Isopropenyl bromide (12.6 ml, 142 mmol) and Et₂O (120 ml) were injected. After small cakes of lithium (2.07 g, 298 mmol) had been added to the suspension, it was irradiated ultrasonically at 0°C for 1.5 h. Most part of the solvent was removed under reduced pressure, then toluene (20 ml) was injected and the mixture was left standing for 20 min. The supernatant was extracted by syringe and diisopropenylzinc was sublimed under reduced pressure of approximately 0.5 mmHg at 40°C. The colorless solid of diisopropenylzinc (3.62 g, 24.5 mmol) was dissolved in an appropriate volume of toluene (22 ml) to give 1.11 N toluene solution of diisopropenylzinc.

Table 2
Enantioselective isopropenylation of various aldehydes



Entry ^a	R	Chiral catalyst	Time (h)		Yield (%)	E.e. (%) ^b
1	Ph	(<i>S</i>)- 1	2	3a	91	89
2	Ph	(1 <i>S</i> ,2 <i>R</i>)- 2	2	3a	73	88
3	4-Cl-C ₆ H ₄	(<i>S</i>)- 1	3	3b	67	87
4	4-Cl-C ₆ H ₄	(1 <i>S</i> ,2 <i>R</i>)- 2	2	3b	84	89
5	4-MeO-C ₆ H ₄	(<i>S</i>)- 1	3	3c	50	82
6	4-MeO-C ₆ H ₄	(1 <i>S</i> ,2 <i>R</i>)- 2	2	3c	68	92
7	1-Naphthyl	(1 <i>S</i> ,2 <i>R</i>)- 2	2	3d	85	79
8	PhCH=CH	(1 <i>S</i> ,2 <i>R</i>)- 2	2	3e	75	73
9	PhCH ₂ CH ₂	(1 <i>S</i> ,2 <i>R</i>)- 2	2	3f	81	73

^a Molar ratio of aldehyde: (*S*)-**1**: diisopropenylzinc was 1.0:0.05:3.0. The molar ratio of aldehyde:(1*S*,2*R*)-**2**:diisopropenylzinc was 1.0:0.20:3.0.

^b E.e. was determined by HPLC using a chiral column.

Typical experimental procedure for the enantioselective isopropenylation: to a toluene solution (2 ml) of (*S*)-DPMPM **1** (6.7 mg, 0.025 mmol) or (1*S*,2*R*)-DIBNE **2** (26.3 mg, 0.10 mmol) was added a 1.11 M toluene solution of diisopropenylzinc (1.35 ml, 1.5 mmol) at 0°C. A toluene solution (2 ml) of aldehyde (0.50 mmol) was added to the mixture. After the reaction mixture was stirred for 2 h at 0°C, the reaction was quenched by addition of saturated aqueous NH₄Cl (10 ml), then the mixture was filtered using Celite and the filtrate was extracted with dichloromethane. The combined extract was dried over anhydrous sodium sulfate and evaporated to dryness under a reduced pressure. Purification of the crude residue on silica gel thin-layer chromatography (TLC) gave the pure allyl alcohol **3**. The optical purity was determined by HPLC analysis using a chiral column.

3.1. (*S*)-2-Methyl-1-phenyl-2-propen-1-ol (**3a**)

IR and ¹H-NMR spectra were in agreement with those in the literature [13]. The enantiomeric excess was determined by HPLC analysis using a chiral column (Daicel Chiralcel OB-H: 4 × 250 mm, 254 nm UV detector, room temperature, eluent: 3% 2-propanol in hexane, flow rate: 1.0 ml min⁻¹, retention time (min) 8 for *R* isomer, 10 for the *S* isomer). Colorless oil; [α]_D²³ = 27.4 (*c* 4.5, CHCl₃) for 89% e.e. The absolute configuration was determined to be *S* isomer by the new Mosher method [14] and the sign of optical rotation coincided with that of (*S*)-**3a** [13].

3.2. (*S*)-1-(4-Chlorophenyl)-2-methyl-2-propen-1-ol (**3b**)

The enantiomeric excess was determined by HPLC analysis using a chiral column (Daicel Chiralcel OB-H:

4 × 250 mm, 254 nm UV detector, room temperature, eluent: 3% 2-propanol in hexane, flow rate: 1.0 ml min⁻¹, retention time (min) 9 for *R* isomer, 11 for the *S* isomer). Tentative assignment of absolute configuration was based on the asymmetric isopropenylation from the *Si* face of benzaldehyde by use of (*S*)-**1** (Table 1, Entry 1). Colorless oil; [α]_D²⁴ = 7.4 (*c* 1.6, CHCl₃) for 89% e.e.; ¹H-NMR δ = 1.59 (s, 3H), 1.95 (s, 1H), 4.96 (s, 1H), 5.13 (s, 1H), 5.19 (s, 1H), 7.31 (s, 4H); IR (neat) 3367, 1651, 1593, 1489, 1442, 906 cm⁻¹; HRMS *Anal.* Found: *m/z* 182.0495. Calc. for C₁₀H₁₁ClO: 182.0498.

3.3. (*S*)-1-(4-Methoxyphenyl)-2-methyl-2-propen-1-ol (**3c**)

The enantiomeric excess was determined by HPLC analysis using a chiral column (Daicel Chiralcel OB-H: 4 × 250 mm, 254 nm UV detector, room temperature, eluent: 3% 2-propanol in hexane, flow rate: 1.0 ml/min, retention time (min) 21 for *R* isomer, 37 for the *S* isomer). Tentative assignment of absolute configuration was based on the asymmetric isopropenylation from the *Si* face of benzaldehyde by use of (*S*)-**1** (Table 1, Entry 1). Colorless oil; [α]_D²⁴ = 51.5 (*c* 1.6, CHCl₃) for 92% e.e.; ¹H-NMR δ = 1.60 (s, 3H), 1.89 (s, 1H), 3.89 (s, 3H), 4.95 (s, 1H), 5.08 (s, 1H), 5.21 (s, 1H), 6.88 (d, 2H, *J* = 7.7 Hz), 7.28 (d, 2H, *J* = 7.7 Hz); IR (neat) 3421, 1651, 1612, 1508, 1454, 903 cm⁻¹; HRMS *Anal.* Found: *m/z* 178.0989. Calc. for C₁₁H₁₄O₂: 178.0994.

3.4. (*S*)-2-Methyl-1-(1-naphthyl)-2-propen-1-ol (**3d**)

The enantiomeric excess was determined by HPLC analysis using a chiral column (Daicel Chiralcel OD: 4 × 250 mm, 254 nm UV detector, room temperature, eluent: 3% 2-propanol in hexane, flow rate: 1.0 ml

min^{-1} , retention time (min) 22 for *S* isomer, 46 for the *R* isomer). Tentative assignment of absolute configuration was based on the asymmetric isopropenylation from the *Si* face of benzaldehyde by use of (*S*)-**1** (Table 1, Entry 1). Colorless oil; $[\alpha]_{\text{D}}^{25} -19.3$ (*c* 1.5, CHCl_3) for 79% e.e.; $^1\text{H-NMR}$ $\delta = 1.67$ (s, 3H), 2.07 (s, 1H), 5.10 (s, 1H), 5.31 (s, 1H), 5.85 (s, 1H), 7.45–8.21 (m, 7H); IR (neat) 3367, 1651, 1604, 1508, 1442, 903 cm^{-1} ; HRMS *Anal.* Found: m/z 198.1043. Calc. for $\text{C}_{14}\text{H}_{14}\text{O}$: 198.1045.

3.5. (*R*)-2-Methyl-5-phenyl-1,4-pentadien-3-ol (**3e**)

IR and $^1\text{H-NMR}$ spectra were in agreement with those in the literature [11]. The enantiomeric excess was determined by HPLC analysis using a chiral column (Daicel Chiralcel OD-H: 4×250 mm, 254 nm UV detector, room temperature, eluent: 3% 2-propanol in hexane, flow rate: 1.0 ml min^{-1} , retention time (min) 25 for *S* isomer, 44 for the *R* isomer). Tentative assignment of absolute configuration was based on the asymmetric isopropenylation from the *Si* face of benzaldehyde by use of (*S*)-**1** (Table 1, Entry 1).

3.6. (*R*)-2-Methyl-5-phenyl-1-penten-3-ol (**3f**)

IR and $^1\text{H-NMR}$ spectra were in agreement with those in the literature [15]. The enantiomeric excess was determined by HPLC analysis using a chiral column (Daicel Chiralcel OD-H: 4×250 mm, 254 nm UV detector, room temperature, eluent: 3% 2-propanol in hexane, flow rate: 1.0 ml min^{-1} , retention time (min) 17 for *S* isomer, 23 for the *R* isomer). Tentative assignment of absolute configuration was based on the asymmetric isopropenylation from the *Si* face of benzaldehyde by use of (*S*)-**1** (Table 1, Entry 1). Colorless oil; $[\alpha]_{\text{D}}^{25} 13.1$ (*c* 1.2, CHCl_3) for 39% e.e.

3.7. 2,3-Dimethyl-1-phenyl-2-buten-1-ol (**4**)

The enantiomeric excess was determined by HPLC analysis using a chiral column (Daicel Chiralcel OD-H: 4×250 mm, 254 nm UV detector, room temperature, eluent: 3% 2-propanol in hexane, flow rate: 1.0 ml min^{-1} , retention time (min) 8 for the major isomer, 11 for the minor isomer). Colorless oil; $[\alpha]_{\text{D}}^{25} 93.0$ (*c* 1.4,

CHCl_3) for 69% e.e.; $^1\text{H-NMR}$ $\delta = 1.50$ (s, 3H), 1.73 (s, 3H), 1.77 (s, 1H), 1.90 (s, 3H), 5.86 (s, 1H), 7.23–7.34 (m, 5H); IR (neat) 3356, 1664, 1601, 1493, 1450 cm^{-1} ; HRMS *Anal.* Found: m/z 176.1201. Calc. for $\text{C}_{12}\text{H}_{16}\text{O}$: 176.1201.

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